UNIVERSITY OF CALGARY

DATA 601 PROJECT

Can We Predict Tomorrow's Economy?

Forecasting Next-Year GDP Per Capita with Machine Learning

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1. About the Data

Data Source: World Bank — World Development Indicators (WDI)

193
Countries

2006–2021
Analysis period

22
Selected indicators

GDP_{t+1}
Prediction target

Country Name	Series Name	2006 [YR2006]	2007 [YR2007]	2008 [YR2008]	2009 [YR2009]
Afghanistan	GDP growth (annual %)	5.357403247	13.82631955	3.92498382	21.39052839
Afghanistan	GDP per capita, PPP (constant 2021 international $)	1929.723897	2155.353068	2191.504356	2565.022086
Afghanistan	GDP per capita growth (annual %)	1.132485055	11.69230329	1.677279156	17.04389632
Afghanistan	Energy imports, net (% of energy use)				
Afghanistan	Energy intensity level of primary energy (MJ/$2021 PPP GDP)	1.5	1.54	1.95	2.27
Afghanistan	Renewable energy consumption (% of total final energy consumption)	31.9	28.8	21.2	16.5
Afghanistan	Access to electricity (% of population)	33.5	38.4	42.4	48.3
Afghanistan	Industry (including construction), value added (% of GDP)	28.21076807	26.88224161	26.915628	21.89712221
Afghanistan	Manufacturing, value added (% of GDP)	16.3855372	17.74730953	17.8391163	13.14987661

Indicator Groups

Economic
GDP • inflation • trade • capital formation

Energy
energy use • electricity access • energy intensity

Industry & Infrastructure
manufacturing • electricity measures

Development & Environment
urbanization • CO_2 • country characteristics

2. Data Preprocessing

Preparing a consistent country-year dataset before EDA and machine learning

1

Reshape

Wide → country-year

Each row = one country in one year

→

2

Select Years

2006–2021

The missing percentage increases sharply after 2021 (>20%)

→

3

Filter GDP Data

217 → 193 countries

Remove countries missing GDP per capita OR GDP growth

→

4

Missing Predictors

Keep NA for EDA

Impute later inside the ML pipeline

Why remove these 24 countries?

GDP is used to construct the ML prediction target, so complete GDP per capita and GDP growth history is required.

Country filtering

(3,472, 26)

shape before

384

rows removed

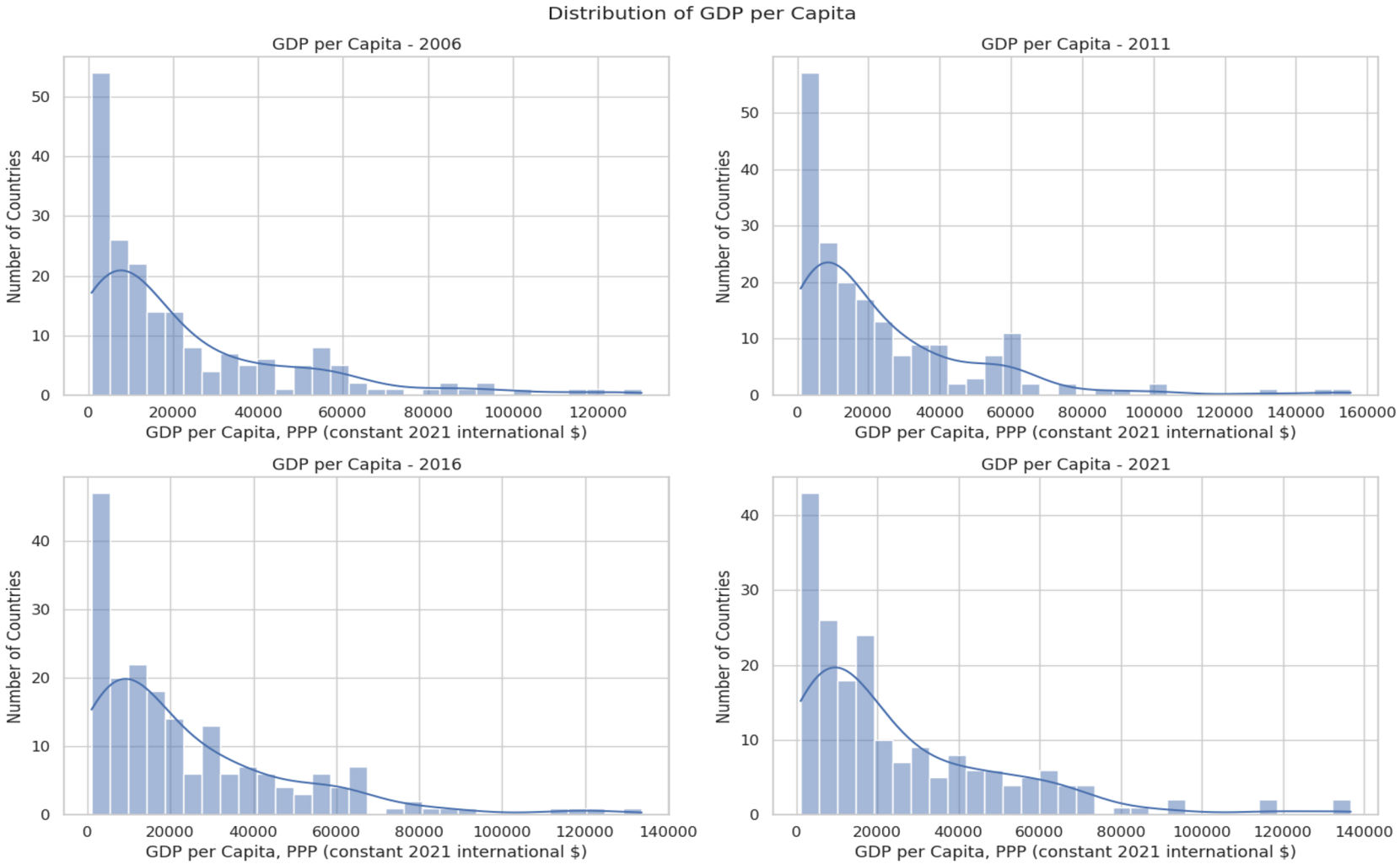
11.06%

of rows removed

(3,088, 26)

shape after

3. Exploratory Data Analysis



GDP per capita is strongly right-skewed

4. Model & Metric Selection

Supervised regression: Year t indicators → continuous GDP per capita in Year t+1

Linear Regression

WHY Simple, interpretable baseline

HOW Weighted linear combination

Decision Tree

WHY Capture nonlinear relationships

HOW Recursive predictor splits

Random Forest

WHY Stable ensemble; reduce tree overfitting

HOW Many trees → average predictions

XGBoost

WHY Powerful nonlinear boosted ensemble

HOW Sequentially correct prior errors

How we evaluate the models

MAE ↓ PRIMARY

Average absolute error in GDP-per-capita units

RMSE ↓ SUPPORTING

Penalizes large prediction errors more strongly

R^2 ↑ SUPPORTING

Proportion of variation in GDP explained

Selection strategy: compare 4 baselines → pick best performance model → tuning → select final model

5. Machine Learning Data Preparation

1 Create Forecast Target

Year t predictors → GDP per capita at t+1

Example: 2019 indicators → predict 2020 GDP

2 Chronological Split

Train: 2006–2015 = 10 years → 66.7%
Validation: 2016–2017 = 2 years → 13.3%
Test: 2018–2020 = 3 years → 20%

3 Encode Country

One-Hot Encoding

Include country information without numeric ranking

4 Handle Missing Predictors

Missing Indicator + Forward Fill + Training Median + Leakage Control

44 inputs entering feature selection = 24 original + 20 missing indicators

5 Feature Selection: Compare Top-Ranked Feature Sets

Chronological validation using predictor years 2016–2017

Feature set	MAE ↓	RMSE ↓	R^2 ↑
Top 3	666.04	1,325.12	.9974
Top 5	642.86	1,278.51	.9976
Top 10	**582.29**	**1,216.47**	**.9978**
Top 15	618.88	1,323.24	.9974
Top 20	623.71	1,276.65	.9976

→ Top 10 selected Lowest MAE = 582.29

Missing-Indicator Check

Top 10 only	MAE 626.29
Top 10 + flags	MAE 581.39
RMSE = 1,201.66	R^2 = .9978

→ Keep 7 missing indicators

Final Feature Reduction

44 VARIABLES

24 original + 20 missing indicators

↓

17 VARIABLES

10 original + 7 missing indicators

27 fewer inputs • 61.4% reduction

6. Model Performance: Baseline → Tuning → Optimization

1 BASELINE MODELS

Compare all 4 models on validation data

Model	MAE ↓	RMSE ↓	R^2 ↑
Decision Tree	948.53	2,233.23	.9925
Random Forest ✓	590.61	1,228.36	.9977
Linear Regression	865.09	1,512.66	.9966
XGBoost ✓	693.83	1,518.44	.9965

SELECT FOR NEXT STAGE

Random Forest + XGBoost

Bagging vs. Boosting

2 AFTER TUNING

Optimize the two selected ensembles

Random Forest

Best parameters = baseline settings

MAE $590.61

RMSE $1,228.36

→ No material improvement

XGBoost

200 trees • depth 5 • LR .05 • subsample 1.0

MAE $601.22

RMSE $1,245.98

→ MAE improved by $92.61

3 AFTER OPTIMIZATION

Test log-transformed GDP target

Random Forest + Log

MAE $604.34

RMSE $1,257.66

→ Log target did not improve RF

XGBoost + Log

MAE $595.45

RMSE $1,221.49

→ Log target improved XGBoost

Train–Validation gap ≈ $180.11

FLOW 4 Baselines → Select RF + XGBoost → Tune Both → Test Log Target → Select XGBoost + Log for better generalization

7. Final Model Selection and Training

Final Model Setup

XGBoost + Log GDP

Tuned parameters

200 trees
max_depth = 5
learning_rate = 0.05
subsample = 1.0

Final training

Train + validation combined
2006–2017

↓

Frozen model

Test once: 2018–2020

Performance

	MAE ↓	RMSE ↓	R^2 ↑
Train 2006–2017	$426.45	$841.13	0.9989
Test 2018–2020	$1,476.45	$3,753.32	0.9782
Naive benchmark	$1,399.39	$3,837.94	0.9772

Generalization gap MAE gap = $1,050.00

What does it mean?

1 High test R^2

The model still explains 97.82% of test variation.

2 Error increased

Test MAE is 3.46× train MAE → generalization is weaker.

3 Benchmark matters

XGBoost has higher MAE than naive, but slightly lower RMSE and higher R^2.

Key takeaway

Strong fit ≠ clear improvement over a simple GDP baseline.

8. Error Analysis: Where Does the Model Struggle?

1 Error by Predictor Year

Predictor year → forecasts the following year

2018 → 2019 — $520

Median % err: 1.6%

2019 → 2020 — $2,350

Median % err: 6.9%

2020 → 2021 — $1,560

Median % err: 4.0%

2019 → 2020 is the hardest period

MAE ≈ 4.5× the 2018 level

2 What Drives Error?

Size of GDP change

Smallest-change group	$532 MAE • 1.6%
Largest-change group	$3,449 • 11.0%

→ Sudden changes are much harder to predict

GDP level

Lowest quartile	$136 MAE
Highest quartile	$3,373 MAE

Median % errors remain roughly

2.5%–5.2% across quartiles

→ **Richer economies have larger dollar errors**

3 Largest Errors

Largest single error

Macao SAR, China

Predictor year: 2019

Current GDP	$133,454
Actual next year	$59,663
Predicted	$128,832
Absolute error	$69,169

Actual GDP change: −55.3%

Error concentration

Top 10 countries → 29.2% of total error

Top 5% observations → 33.9%

Main finding: the generalization gap is concentrated in sudden economic changes—especially the 2020 shock and subsequent rebound.

THANK YOU

for listening

Q & A

Questions • Comments • Discussion

We'd be happy to hear your questions.

Appendix 1. Feature Selection

1 Permutation Importance

Training-period ranking • Country excluded

Top 10 ranked predictors

1. Current GDP per capita	24,167.8
2. Inflation	50.1
3. Electric power consumption	18.3
4. Energy use per capita	17.3
5. CO_2 emissions per capita	16.9
6. Urban population	14.0
7. GDP per capita growth	11.7
8. Renewable energy consumption	9.2
9. Internet use	7.3
10. Energy intensity	7.0

Value = mean increase in MAE after shuffling

2 Chronological Validation

5-Fold Expanding-Window Validation

Train on past → validate on next period

Feature set	5-Fold Avg. MAE
Top 3	$666.04
Top 5	$642.86
Top 10	**$582.29**
Top 15	$618.88
Top 20	$623.71

Selected: Top 10

Lowest average MAE across 5 folds

3 Final Input Checks

Missing indicators

Top 10 only	**MAE $626.29**
Top 10 + missing flags	**MAE $581.39**

→ Keep 7 related missing flags

Country sensitivity check

Compare the same selected predictors:

WITHOUT Country ↔ WITH Country

Country tested separately from the permutation ranking.

Final numeric inputs: 10 + 7 flags = 17

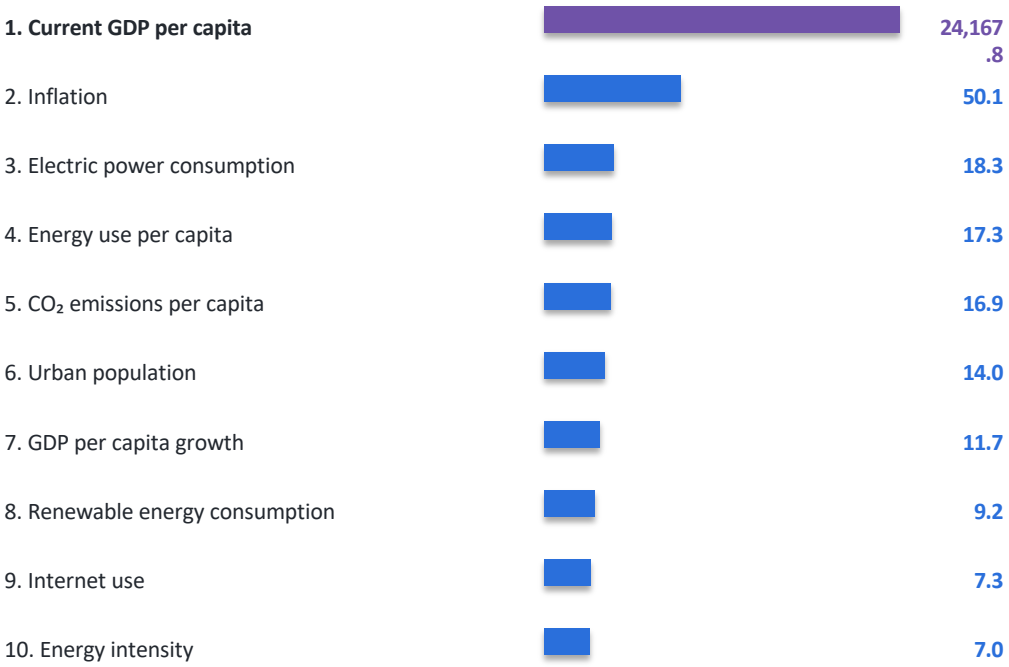
Result: Top 10 predictors had the lowest 5-fold average MAE; adding missing indicators improved performance further.

Appendix 2. What Drives Next-Year GDP Predictions?

Interpreting feature contributions — not just model accuracy

Permutation Importance Ranking

Increase in MAE when each feature is shuffled



What does this tell us?

1 GDP persistence dominates

Current GDP carries far more predictive information than any other feature.

2 Other indicators add smaller signals

Inflation, energy/electricity use, emissions, urbanization and GDP growth contribute additional information.

3 Importance ≠ causal effect

Permutation importance tells us how useful a feature is for prediction — not whether it causes GDP to rise or fall.

Connection to benchmark This dominance helps explain why the naive GDP benchmark was so difficult to beat.

Main insight: short-term GDP forecasting is driven primarily by GDP persistence; other indicators provide incremental predictive information.